Wenrui **Zhong**

Room 1223, Building 9, 8 Nobel Way, M1 7FA

07442667382 | wenrui.zhong24@imperial.ac.uk | Wenrui Zhong

Education

Imperial College London *London, UK*

Master of Advanced Aeronautical Engineering *09/2024-06/2025(expected)*

- Courses: Advanced Propulsion, Aerothermodynamics of Launchers and Re-entry Vehicles, Orbital Mechanic, Spacecraft System

The University of Manchester *Manchester, UK*

Bachelor of Science (Honours) in Physics with Astrophysics *09/2021-06/2024*

- CGPA: 75.9/100.0
- Relevant Courses: Introduction to Astrophysics and Cosmology, Special Relativity, Astrophysical Processes, Electromagnetism , Astronomical Plasma with Nuclear Fusion, Fluid Mechanics and Physics of the Solar System, Lagrangian Dynamics

D'overbroecks College *Oxford, UK*

IGCSEs and A-levels *09/2018-06/2021*

- IGCSEs: 9 in Math and Physics, 8 in Further Math and Chemistry
- A levels: Math(A*), Further Math (A*), Physics(A*)
- Achievements: Merit in British Mathematics Olympiad; Merit in Kangaroo Competition; Gold in Senior Math Challenge in Year 11; Upper Bronze in British Physics Olympiad.

Skills

Languages Chinese (Native), English (Fluent), Cantonese (Fluent)

Software Python, Julia, Matlab, Spider, DS9, Maerstro, Latex, Google Documents, and MS Office.

Lab Experience

Higgs Boson Experiment (8 Weeks) *Manchester, UK*

02/2024 - 05/2024

- Predicted the background photon noise in ATLAS experiment with multiple statistical model by histogram analysis using ROOT and Python.
- Verified the existence of Higgs boson from difference between statistical noise prediction and actual data through significance calculation.
- Conducted calculation foe the cross section of Higgs boson and compare the result with theoretical prediction and justified the statistical model.

Crab Pulsar Experiment (8 Weeks) *Manchester, UK*

9/2023- 12/2023

- Operated the 42-ft radio telescope at Jodrell Bank Observatory on-site, observing and recording pulses of various pulsars over eight weeks.
- Developed Python programs to calculate pulsar dispersion measures, determine pulsar periods using Fourier analysis, and analyze the Crab pulsar's period evolution.

Lab Experiments Series (200 Hours) *Manchester, UK*

09/2021- 06/2023

- Covered various topics including astrophysics, particle physics, circuits and electromagnetism.
- Ultrasonics by Magnetostriction Experiment: Determined the speed of sound and attenuation coefficient of nickel after applied magnetostriction, and used both the continuous wave and pulse methods.
- Galactic Hydrogen: Used the 6.4-meter radio telescope in Jodrell Bank to analyse the galactic plane, constructed the flat rotational curve for a galaxy, and predicted the existence of dark matter.
- Properties of Gamma Radiation: Operated a scintillator and measured Na-22 source with sum peak method, and compared the absorption coefficient for gamma-ray with different absorber at different gamma energies.

Academic Projects

Undergraduate Dissertation on Low Radio Interferometry on Lunar Farside

Manchester, UK

02/2024 - 05/2024

- Discussed the environment of lunar far side and evaluated optimal location for stable temperature and least radio interference to conduct radio interferometry.
- Constructed multiple models for the structure of interferometry and analyzed the optimal structure for the observation on early universe with Python.
- Researched 28 papers discussed about the radio interfeometry and lunar far side and developed skills and experiences as researcher.

General Physics Skill Group Project on Neutrino Communication with Alien

Manchester, UK

02/2024 - 05/2024

- Researched the properties of neutrino and Estimated the background noise of neutrino due to star emission.
- Calculated the minimum power required of the neutrino signal for an alien civilization to receive.
- Learned to lead a group project and constructed different tasks for each member of the team.

Programming for Physicists Course Projects with Pythom

Manchester, UK

09/2022 - 01/2023

- Bouncy Ball: Wrote a program capable of making a series of calculations, such as bouncing time regarding a bouncy ball bouncing above some minimum height.
- 79Rb Decay: Determined the half-lives and decay constants of both 79Sr and 79Rb from activity data, obtained the uncertainties on these values, and produced contour plot to support analysis.

Extracurricular Activity

Imperial College London Space Society

Imperial College London

Member

10/2024-06/2025

- Joined the Orbital division in Project Svarog which are planning to send the first student-lead solar-sail to outer solar system.
- Constructed the low earth orbit launch simulation for 2025 with teammates and acquired the experience of Julia coding and teamwork experience.

Astronomy Society

The University of Manchester

Member

09/2021 - 06/2024

- Joined multiple events, for example, a stargazing trip and water rocket workshop, and built a rocket powered by pressurized gas in a water bottle.

Undergraduate Summer School of National Astronomical Observatories

The University of Chinese Academy of Sciences

Member

07/2024-07/2024

- Attended multiple lectures on deep space observatories deployment such as Einstein Probe which demonstrated the function and necessity of these missions
- Discussed with several researchers and expanded my understanding on deep space exploration.

Undergraduate Summer Camp of National Astronomical Observatories

The University of Chinese Academy of Sciences

Member

07/2023-07/2023

- Attended multiple talks on solar observations and high energy radiation.
- Viewed and learned the function of multiple telescope at Yunnan such as Yunnan 40-meter radio telescope

PASS Leader

The University of Manchester

PASS

09/2022 - 06/2023

- Collaborated with 2 other PASS Leaders to provide academic and university life guidance to a group of first- year students.
- Gained experience in peer tutoring and mentoring.

Special Topic Leaflet Group

The University of Manchester

Member

10/2021-11/2021

- Created a scientific leaflet on Pulsars, covering the history of pulsars and their properties.
- Distributed these leaflets among high school students to arouse their interest in Physics.

United Kingdom
Mathematics Trust

British Mathematical Olympiad
Round One 2020-21

Alan Zhong

of

d'Overbroeck's

is awarded a

CERTIFICATE OF MERIT

[XTX] MARKETS

The UKMT's work is generously supported by XTX Markets.
XTX is a world-leading algorithmic trading firm delivering
quality liquidity across financial markets.

Graham Keniston-Cooper
Chair, UK Mathematics Trust

United Kingdom
Mathematics Trust

Senior
Mathematical Challenge 2020

Alan Zhong

of

d'Overbroeck's

is awarded a

GOLD CERTIFICATE

BEST IN SCHOOL

[XTX] MARKETS

The UKMT's work is generously supported by XTX Markets.
XTX is a world-leading algorithmic trading firm delivering
quality liquidity across financial markets.

Graham Keniston-Cooper
Chair, UK Mathematics Trust

United Kingdom
Mathematics Trust

Senior Mathematical Challenge 2019

Alan Zhong

of

d'Overbroeck's

is awarded a

GOLD CERTIFICATE

Graham Keniston-Cooper
Chair, UK Mathematics Trust

BPhO
British Physics Olympiad

Physics Challenge

This certificate is awarded to

Alan - Zhong Wenrui

who achieved a

Bronze (upper level)

in the British Physics Olympiad 2018-19
Physics Challenge competition

Robin Hughes

Mr Robin Hughes
Chair of the British Physics Olympiad
March 2019

United Kingdom
Mathematics Trust

Senior Mathematical Challenge
2018

WENRUI (ALAN) ZHONG

of

D'OVERBROECK'S

is awarded a

GOLD CERTIFICATE

Institute
and Faculty
of Actuaries

Professor Chris Budd OBE
Chairman, UK Mathematics Trust